Numbers Through Signs: How LLMs Connect Counts, Number Words, and Numerals

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Abstract

Large language models manipulate numerical expressions fluently, yet it is unclear whether a number inferred from a collection enters the same internal interface as an explicit numeral or number word. We test digits, English number words, and counts of repeated tokens in five Qwen3 base models from 0.6B to 14B. After each quantity expression but before a comparison, parity, or equality question, we interchange one residual-stream state between forms. Textual-count states preserve held-out cardinal order, even at fixed sequence length. In behaviourally competent 8B and 14B cases, digit and number-word states are bidirectionally interchangeable. At the tested collection boundary, however, a count state does not transfer even to a different count of the same form, and no count–sign direction meets the same one-shot criteria. A same-depth sweep and symbol-derived linear subspace do not recover a bidirectional local interface. Longer interventions and a separate mixture-of-experts model reveal strong but directional bridges. Counting information and correct behaviour can therefore coexist without a locally portable state at the tested boundary.

1 Introduction

Large language models (LLMs) can manipulate equations, check proof steps, and answer many mathematical questions, while remaining unexpectedly brittle at counting repeated items or extracting quantities from ordinary text. Recent mechanistic work complicates any simple explanation based on missing number information. Counts can be decoded from language- and vision-language-model activations, and multi-stage routines can transmit and aggregate counting signals (Hasani et al., 2026b,a). A model may therefore contain a count even when its answer is unreliable. The open issue is how that count participates in the rest of the model's numerical computation.

Related studies find regular geometry across explicit number forms (Štefánik et al., 2026), and show that numerical information recoverable by a probe can exceed what a model expresses in its answers (Yuchi et al., 2026). Low-dimensional, format-agnostic subspaces can also transfer reasoning more effectively than replacing an entire hidden state (Yuan and Su, 2026). These results establish numerical structure and usable signals. They leave a sharper question unanswered: once the model has constructed the cardinality of a collection, can that state be used in place of the state produced by a digit or number word in a later operation?

The question matters because human numerical expressions are ordinarily reused across very different encounters. The mark *3*, the word *three*, three objects, and three repeated actions can all guide a new comparison, match, or calculation. Humans develop these abilities through language embedded in perception, action, and cultural practice. A text-first LLM follows a strikingly different route: proofs, numerals, descriptions of objects, and reports of action all reach it through the same stream of signs. This makes the model an inspectable case for separating fluency with public mathematical notation from the internal connection between signs and constructed quantities.

We study that connection with a task-late interchange assay. A Qwen3 model first reads a quantity as a digit, number word, or enumerated textual collection. Only after the expression ends does it receive a comparison, parity, or equality question. We transplant the boundary state from one form into another and ask whether the model's answer follows the donor number. The experiments show a robust digit–word interface, but the tested collection-boundary state is not portable even between two textual counts. Partial transfer under longer interventions shows that the count–sign connection is graded rather than absent. For mathematical NLP, the result separates what behavioural ac-

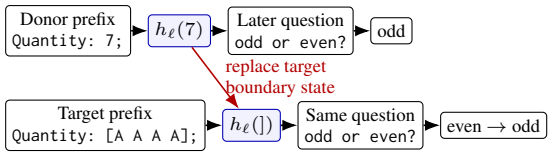


Figure 1: **One-shot task-late interchange.** The donor encodes seven as a digit; the target encodes four as a textual count. Both prefixes end before the parity question. If the transplanted state can play the donor's numerical role, the target answer changes from even to odd. A same-number donor tests whether cross-form replacement preserves the original answer.

curacy and numerical probes measure from whether different expressions feed a common, immediately reusable numerical state.

2 Task-Late Interchange Assay

Models, number forms, and tasks. We study base Qwen3 checkpoints at 0.6B, 1.7B, 4B, 8B, and 14B parameters (Yang et al., 2025). Production runs use pinned revisions, BF16 weights and activations, and one A100 80GB GPU, without quantization or CPU offload. A Qwen3-30B-A3B checkpoint is analysed separately because sparse expert routing makes it an architectural contrast rather than a sixth dense scale point. Cardinalities 2–9 are expressed as an Arabic digit (`3`), an English number word (`three`), a variable-length textual count (`[A A A]`), or a shuffled ten-item A/B sequence containing three A tokens. The last condition keeps total length constant while cardinality changes. The later questions ask whether the quantity exceeds a threshold, is odd or even, or equals a threshold.

Interchange by example. Figure 1 shows the central intervention. The donor prefix states seven explicitly; the target prefix presents four repeated tokens. At one decoder layer, we replace the residual-stream state at the target expression's closing bracket with the donor state at the digit. The unmodified target answers *even*; successful counterfactual transfer makes the continuation follow seven and answer *odd*. Replacing the boundary state with one derived from a different form of the same number should preserve the target's answer. Together, counterfactual transfer and same-number preservation test *local causal interoperability*: whether one task-late state can assume another form's numerical role in the model's own downstream computation (Geiger et al., 2025).

Selection, screening, and measures. For each model, the decoder site is selected using development-template digit–word transfer only; the embedding output is excluded. The selected sites for the dense series are 3, 4, 6, 6, and 1 in ascending model size. Evaluation wording is chosen through behaviour alone: digit and word balanced accuracy must each reach .85, textual-count accuracy .65, each count answer class must exceed .60, and every principal direction must retain at least 16 trials on which the unmodified donor and target are correct. Conditional interchange intervention accuracy (IIA) is the proportion of those eligible trials whose patched answer follows the donor. A same-number donor supplies a do-no-harm preservation test; exact no-op and wrong-donor trials check the intervention itself. A direction is counted as established when eligibility is at least .50, same-number preservation .90, one-shot IIA .80, and the lower cluster-bootstrap bound on donor-directed logit gain is positive.

Persistence, readouts, and local scope. The primary intervention is applied once; additional windows maintain the donor state over 2, 4, 8, or all remaining decoder sites. For 14B comparison, we also begin a to-end intervention at every same-depth site. Ridge readouts leave out one numerical value at a time, and a rank-7 cardinal subspace is learned only from digit and word development states. The assay targets one local interface, not an information bottleneck: after the closing-bracket state is replaced, later tokens can still attend to the original A tokens.

3 Results

Collection states preserve cardinal order. A ridge readout trained on textual-count states orders unseen values with Spearman $\rho = .990$–$.993$ across all five dense models (Fig. 2a). The fixed-total A/B control reaches $\rho = .967$–$.993$, so total sequence length is unnecessary for this within-format result. A digit-trained readout transfers to number words ($\rho = .891$–$.956$). Applied to textual counts, however, the same readout gives $\rho = -.707$ to $-.967$. The observed rank reversal is directional: digit-trained readouts also give $\rho = -.775$ to $-.922$ on fixed-total collections. Varying sequence length therefore does not explain this systematic misalignment of the tested linear coordinates, whose origin remains unresolved.

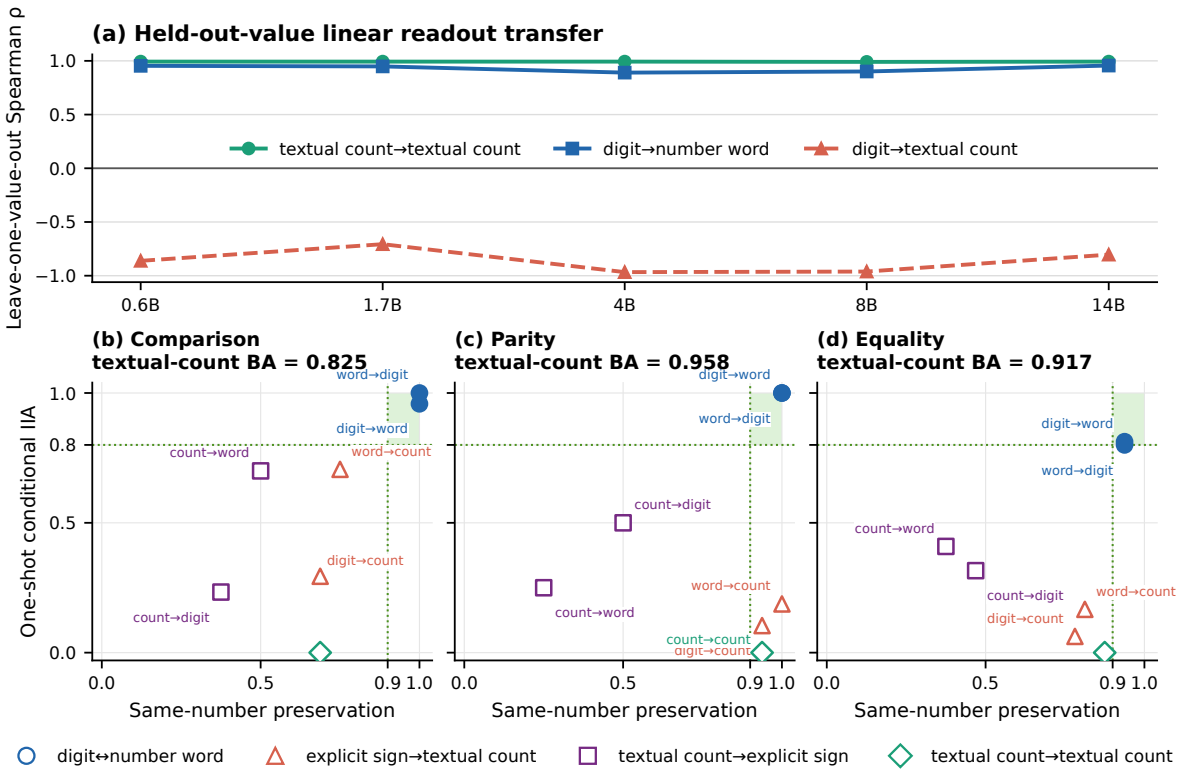


Figure 2: **Cardinal readouts and one-shot interchange.** (a) A readout trained on textual counts preserves held-out numerical order; a digit-trained readout transfers to number words but not to textual counts. The fixed-total control reaches $\rho = .967$–$.993$ within format. (b–d) Same-number preservation and conditional IIA for Qwen3-14B on the three behaviourally screened tasks, including the previously unreported textual-count-to-textual-count baseline. Filled points satisfy the four stated one-shot criteria.

Digits and number words provide the causal anchor. Qwen3-14B passes behavioural screening on textual counts for comparison (.825 balanced accuracy), parity (.958), and equality (.917). Across all three operations, digit and number-word states meet the one-shot criteria in both directions. Comparison and parity have same-number preservation of 1.000 and IIA of at least .958 in each direction; equality has preservation of .938 and IIA of .800–.813. The positive anchor demonstrates that the assay can detect a locally portable state shared by two expression forms.

Competent counts do not show the same one-shot interface. Parity supplies the clearest contrast because every requested 14B count–sign intervention has correct unmodified endpoints: each direction contains 48 eligible trials, versus 24 for each explicit-sign direction. Digit-to-count replacement preserves a same-number answer at .938 but follows a different donor number on only .104 of trials; number-word-to-count preservation is 1.000 while IIA is .188. The two explicit-sign directions score 1.000 on both measures. The 8B model independently passes an equality wording with textual-count balanced accuracy .813: both digit–word directions meet the criteria, while none of the four count–sign directions does. Across comparison, parity, and equality, zero of twelve 14B count–sign directions establish the one-shot local interface.

The count boundary is not a portable register. The full format matrix supplies a same-form baseline. At the selected 14B site, count-to-count one-shot IIA is zero for comparison, parity, and equality. Parity is decisive: all 48 trials are eligible, and preservation is .938 over 16 same-number trials. Thus the tested closing-bracket state does not carry a counterfactually portable count even within the collection format; the cross-form null is not merely disagreement between two portable codes.

Depth and linear-subspace controls preserve the dissociation. For 14B comparison, to-end replacement beginning at every same-depth site finds digit–word transfer at many starting depths, but no identity-valid bidirectional count–sign relation. A rank-7 subspace learned from digit and

number-word states gives IIA and preservation of 1.000 within the explicit sign pair. For count-to-digit, subspace replacement raises preservation to 1.000 while IIA remains .067. No count direction becomes bidirectional, nor do equal-rank random, label-permuted, or top-variance controls reveal one.

Longer interventions reveal partial bridges. Count–sign replacements frequently move the output margin towards the donor even when the categorical answer does not change. Maintaining the transplanted state can produce much stronger transfer: on 14B parity, number-word-to-count IIA rises from .188 after one site to .729 over eight. The separate 30B-A3B mixture-of-experts model provides a stronger boundary case on one equality wording. Digit-to-count reaches .806 at window 8, and number-word-to-count reaches .833 at windows 2–8, all with same-number preservation 1.000. The reverse directions and a second equality wording are weaker. Text-only training can therefore connect explicit signs to a constructed count, but the strongest connections remain directional and dependent on prompt or continued intervention.

4 Discussion

The local failures have different immediate diagnostics. Count-to-count and sign-to-count results show that the closing bracket is not a causal bottleneck: the model can continue to use the surviving collection tokens, while sustained sign-to-count intervention increasingly competes with that evidence. Full-state count-to-sign replacement can also disrupt an incompatible sign slot. Yet the symbol-subspace count-to-digit intervention preserves the same-number answer perfectly while transferring the donor on only .067 of eligible trials. Behavioural numeracy, probe decodability, causal influence, and immediate cross-format reuse are therefore distinct measurements for mathematical NLP.

The experiments separate properties often collapsed into saying that a model "represents" a number: recoverable information, behavioural use, causal influence, and immediate reuse across expressions. Broader coordination could instead be distributed across states and remain flexible across directions, wordings, and operations. The tested Qwen3 states separate information and influence from this local form of coordination.

For humans, learning that *3* means *three* is more than memorizing a pair of marks: both signs can be used for a quantity already encountered in counting, matching, and acting on collections. Qwen3's digit–word interchange shows that language training can coordinate signs internally. The symbol-grounding problem asks how such signs connect beyond one another (Harnad, 1990); model tokens may also inherit derivative reference from the human practices recorded in their data (Mandelkern and Linzen, 2024). Our dissociation concerns how far that inherited practice organises the model's own collection-derived states.

Embodied accounts suggest why human numerical uses may integrate readily (Lakoff and Núñez, 2000; Winter and Yoshimi, 2020). The body need not store an abstract object called three. It can provide a common causal ground: matching, repetition, distribution, addition, removal, and persistence constrain signs and situations through recurring successes and failures. The familiar *felt threeness* of three objects or actions marks a phenomenological contrast; our experiment tests functional organisation, not experience. Infants distinguish small-set tracking from approximate magnitude systems (Feigenson et al., 2004), while non-human animals act selectively on quantity with limited public symbolism (Nieder, 2021). Text-first LLMs expose the complementary case: extensive public symbolism without a history of quantity-guided action. How nonsymbolic quantity connects to exact number words remains disputed (Leibovich and Ansari, 2016).

The positive results sharpen this contrast. Qwen3-14B coordinates explicit signs tightly, while the MoE model develops strong symbol-to-count transfer under sustained intervention. Such directional, unstable bridges differ from flexible reuse across human numerical practice. Testing embodied development requires changing how numerical forms are learned, not further interpreting one pretrained text-only family.

5 Conclusion

Textual counts carry cardinal information and enter partial causal bridges, but do not share the portable digit–word interface. At the tested boundary in competent 8B and 14B models, a count state transfers neither within its format nor to an explicit sign after one shot. Same-depth and symbol-subspace tests do not restore bidirectional transfer; longer interventions reveal a fragile connection.

Limitations

The five dense checkpoints belong to one model family and do not provide a controlled scaling law. Cardinalities are restricted to 2–9, and every collection is textual. Three comparison, three parity, and four equality wordings were screened; one 14B wording per operation passed the behaviour-only rule. The principal causal results therefore concern prompts with adequate baseline competence rather than prompt variation in general. The cluster-bootstrap intervals summarise variation over eight source values within this finite synthetic design; they do not estimate a population over numbers, prompts, model families, or independently trained checkpoints. The 30B-A3B contrast changes sparse routing, total parameter count, active parameter count, and other properties together.

The intervention replaces one task-late token. The original items in a textual collection remain available through attention, allowing a counterfactual donor state to compete with surviving target evidence. Source and target states are compared only at the same decoder depth. The experiments therefore leave open distributed, multi-token, nonlinear, temporally shifted, and collection-trained interfaces. The all-depth sweep covers same-depth correspondences, while ridge readouts and the rank-7 symbol-derived subspace test linear coordinates learned from specific development templates.

The operations and thresholds measure local causal interoperability rather than grounding, phenomenal experience, or a complete theory of mathematical understanding. No model learned quantity through perception or action. A direct follow-up could train matched agents in a synthetic 2-D environment under three curricula: text-only descriptions, passive views of collections, and action-contingent tasks requiring agents to match, produce, and distribute n objects. Novel signs could be learned in disjoint contexts without direct pairing. Their later bidirectional reuse across new prompts and operations would test whether shared success conditions create the common numerical ground proposed here.

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